



A sentiment corpus for the cryptocurrency financial domain: the CryptoLin corpus

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Abstract

The objective of this paper is to describe Cryptocurrency Linguo (CryptoLin), a novel corpus containing 2683 cryptocurrency-related news articles covering more than a three-year period. CryptoLin was human-annotated with discrete values representing negative, neutral, and positive news respectively. Eighty-three people participated in the annotation process; each news title was randomly assigned and blindly annotated by three human annotators, one in each different cohort, followed by a consensus mechanism using simple voting. The selection of the annotators was intentionally made using three cohorts with students from a very diverse set of nationalities and educational backgrounds to minimize bias as much as possible. In case one of the annotators was in total disagreement with the other two (e.g., one negative vs two positive or one positive vs two negative), we considered this minority report and defaulted the labeling to neutral. Fleiss's Kappa, Krippendorff's Alpha, and Gwet's AC1 inter-rater reliability coefficients demonstrate CryptoLin's acceptable quality of inter-annotator agreement. The dataset also includes a text span with the three manual label annotations for further auditing of the annotation mechanism. To further assess the quality of the labeling and the usefulness of CryptoLin dataset, it incorporates four pretrained Sentiment Analysis models: Vader, Textblob, Flair, and FinBERT. Vader and FinBERT demonstrate reasonable performance in the CryptoLin dataset, indicating that the data was not annotated randomly and is therefore useful for further research¹. FinBERT (negative) presents the best performance, indicating an advantage of being trained with financial news. Both the CryptoLin dataset and the Jupyter Notebook with the analysis, for reproducibility, are available at the project's Github. Overall, CryptoLin aims to complement the current knowledge by providing a novel and publicly available Gadi and Ángel Sicilia (Cryptolin dataset and python jupyter notebooks reproducibility codes, 2022) cryptocurrency sentiment corpus and fostering research on the topic of cryptocurrency sentiment analysis and potential applications in behavioral science. This can be useful for businesses and policymakers who want to understand how

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cryptocurrencies are being used and how they might be regulated. Finally, the rules for selecting and assigning annotators make CryptoLin unique and interesting for new research in annotator selection, assignment, and biases.

Keywords News · NLP · Cryptocurrency · FinBERT · Events · Labeled dataset · Sentiment corpus

1 Introduction

Annotated corpora play a crucial role in facilitating the data-driven development of procedures and machine learning approaches, as highlighted by Poesio et al. (2016). Despite the recent proliferation of corpora resulting from web-crawling, there remains a scarcity of annotated corpora, particularly those that are reliably annotated. This is particularly evident in the field of finance, where there is a pressing need for more comprehensive and reliable annotated corpora (Daudert 2022; Gadi and Ángel Sicilia 2022).

Annotated corpora are particularly valuable in the field of event study. The event study approach for financial decisions has been widely used for the general stock market. Marty et al. (2020) presents a survey of several works applying text analytics to the study of news media in financial markets beyond intraday horizons. The survey shows a great variety of techniques applied to event study ranging from parametric techniques like linear regression (including abnormal return calculated from the Fama–French three-factor model), vector autoregression, volatility models, analytic models, nonparametric techniques, semiparametric and model-free techniques.

In the cryptocurrency space, numerous studies use news for cryptocurrency price prediction. Notable articles include those that use social media (Beck et al. 2019), tweet volumes and sentiment analysis (Abraham et al. 2018), news and social media sentiment (Lamon et al. 2017), sentiment analysis (Khurshid 2017), news articles (Ider 2022), and Elon Musk’s Twitter activity (Ante 2021, 2022). Cryptocurrencies refer to digital assets that can be used as a medium of exchange between individuals or groups. They are outside of the traditional financial system and date back to the first proposal of Bitcoin in 2009 by an anonymous individual or organization known as Nakamoto (2008). Cryptocurrencies are a form of decentralized clearance system that provides anonymous record-keeping and transactions via a public blockchain that anyone can see. A blockchain is a system in which transaction records are maintained across several computers linked by a peer-to-peer network. Cryptocurrencies have become a fruitful area of research interest across many disciplines, from physics and computer science (with studies in graph visualization) Liu et al. (2020) to economics (see Demir et al. 2018).

However, as far as we know, event studies separating the effect generated by an event from the trend/volatility are still scarce and scattered, and in most cases centered on subsets of events and Bitcoin only. Important works in the field include:

- Štefan Lyócsa et al. (2020)—assess how scheduled macroeconomic news announcements affect the volatility of Bitcoin.
- Othman et al. (2019)—assess the effect of symmetric and asymmetric information on the volatility structure of Bitcoin.
- Klein et al. (2019)—assess the effect of a subset of cryptocurrency news related to regulation and regulation-related opinions and rumors centered on Bitcoin.
- Gurrib et al. (2019)—assess the effect of major global macroeconomic news on several cryptocurrency price movements, including Monero.
- Nguyen et al. (2019)—assess asymmetric impacts of tightening/easing monetary regimes on our four major cryptocurrencies, including Bitcoin, with special attention to Chinese events' impact.

There could be various reasons for the small amount of research on event studies in cryptocurrency. These reasons include cryptocurrency remaining a very volatile asset where it is still difficult to clearly isolate the proportion of abnormal generated by events from the ones reflecting its intrinsic volatility. Additionally, there are several structural differences, such as being 24×7 and globally traded, which require specific studies to understand which are the key events that can trigger up a change or conservation of trend. Nevertheless, the decentralized nature of cryptocurrencies may generate a belief that certain events (news) are crypto-related events only (for example, crypto bans in China, the creation of a Bitcoin ETF, the launch of a new blockchain, etc.), and perhaps other general events (terrorism, wars, etc.) may have an effect on crypto markets different than they do on markets in general. Although there are contributions in this regard, they all rely on having a high-quality curated database of events classified by type. Here we depart from the hypothesis that those databases should be specific to the events related to the cryptocurrencies and blockchain domain.

To complement the current resources for event studies for the specifics of cryptocurrency-related research, we describe here CryptoLin, a novel and publicly available cryptocurrency sentiment corpus. It covers more than a 3 year period. In addition, CryptoLin is annotated with a transparent mechanism designed to enable the detection of errors and correction by researchers using the dataset. We have also compared the results of the annotation process with existing sentiment models to obtain some insights into its strong attributes as a complementary dataset. In conclusion, a manually labeled news events corpus for cryptocurrency is an important contribution to the field by providing valuable data for researchers to study market trends and financial decision-making in the cryptocurrency industry. It would also help address the current lack of reliably annotated corpora in the field of cryptocurrency.

The remainder of this paper is organized as follows. First, Sect. 2 discusses related works on sentiment corpus extraction and manual annotation. Section 3 describes our article collection approach used to obtain a corpus representative of cryptocurrency news, the annotator selection criteria (which is a key aspect of this paper), annotated units, and the procedure used for annotation. Section 4 covers the inter-annotator agreement study. In Section 5, we present a preliminary

analysis with several pre-trained algorithms available in the CryptoLin dataset. Finally, in Sect. 6, we conclude with some notes on the availability of the corpus and acknowledgements.

2 Related works

2.1 Methods for sentiment corpus annotation

In 2021, Daudert (2022) points out that two main challenges arise when applying sentiment analysis: relevance and point of view. First, relevance can be summarized as setting up common rules for dropping irrelevant news. Second, point of view refers to finding a common point of view during the annotation process. The second is particularly important for crypto since news regarding crypto mining may be positive for the miners but negative for the environment, generating inconsistent annotation if a point of view is not previously set. The same article also shows that although the two challenges exist, it does not interrupt a myriad of corpora from existing. The article highlights a selection of recent corpora, and we reproduce the first three here:

- IMDB corpus Contains 25,000 textual movie reviews from IMDB annotated as positive and negative (Maas et al. 2011).
- Sentiment stanford sentiment treebank. Comprises reviews from the entertainment review website Rotten Tomatoes annotated using a sentiment scale between 1 and 25 (Socher et al. 2013).
- Sentiment140 Consists of 160,000 tweets annotated in polarities (positive, neutral, negative) after removing emoticons (Go et al. 2009).

Daudert (2022) introduces FinLin, an annotated corpus containing investor reports, company reports, news articles, and microblogs from multiple sources. In this work, the data is annotated using AWOCATo, the author's annotation tool (Daudert 2020). The AWOCATo tool offers a two-dimensional scale annotation of sentiment score and a relevance score, and each annotator labels each text. Having three annotations per text in place, the FinLin dataset was consolidated either automatically or manually by an additional annotator (i.e., consolidator). The consolidation was conducted with AWOCATo, the same tool used for annotation when there were controversial cases. The author presents the detailed hiring process from selecting candidates to selecting the final three annotators with specified profiles:

- Higher education in finance, business, or economics;
- Professional experience related to finance, business, or economics;
- Stock trading experience;
- English native.

Jacobs and Veronique (2022) presents a corpus of fine-grained company-specific events in English economic news articles. In this article, the authors web crawled

news from <https://finance.yahoo.com/>, which works as a business news aggregator. In this work, three annotators used a consensus decision methodology as each document was fully annotated by the three annotators.

Meireles et al. (2021) focused on presenting and proposing a consensus mechanism. The consensus mechanism consisted of taking the consensus of three working groups (as stated in the article, “A modified Delphi process (online or offline survey) was utilized to create a consensus survey based on suggested recommendations from each of the working groups. This was performed in three rounds”). The consensus mechanism was successfully tested in an annotation of video data, where annotators were clinicians, engineers, and data scientists divided into work groups.

Recently, in 2023, Pozzi et al. (2023) proposes a novel methodology for applying news aggregation and summarization techniques to cryptocurrency and blockchain news. Natural language processing solutions were used to cluster news by topic and summarize articles from different newspapers. The methodology was effective in most cases, with 86.8% of summaries being coherent and 95.7% of articles correctly aggregated.

2.2 Existing corpora

In the specific field of Cryptocurrency news, Köse (2020) makes available a very small annotated dataset with 100 hundred news including cryptocurrency descriptions annotated with sector, asset type and transaction anonymity labels. For annotation, the authors mention they have selected randomly one hundred news (100) from RSS feed and requested three experts to annotate news’ title as positive and negative and neutral. The article also points out that they were able to achieve promising results by fine-tuning FinBERT with this 100 news dataset, but as future work, the authors plan to increase this training data by annotating more news.

Thus, CryptoLin aims to complement the current knowledge and available resources by providing a much larger cryptocurrency repository containing 2683 relevant manually annotated news from a set of diverse annotators. This objective is to foster research on the topic of cryptocurrency sentiment analysis and its potential applications in behavioral science.

3 Corpus design

As observed in Daudert (2022), sentiment annotation requires a relevance score as well as the sentiment score itself if data is collected from generic sources. In order to avoid this step, we followed the pipeline set by Jacobs and Veronique (2022) which used a leading news aggregator (in their case: Yahoo! Finance). Jacobs and Veronique (2022) crawled English online business news articles on all companies in the S &P500 over a period of 14 months using the business news aggregator Yahoo! Finance API.

In our case, we have decided to use CoinMarketCal (2022) whose main source of news is The Block, a leading news aggregator. According to their website, The

Block is dedicated to transparent journalism and had 65 employees as of Tuesday, May 3rd, 2022.

Moreover, we have adopted the consensus mechanism proposed by Meireles et al. (2021) using three individual annotators selected from independent and diverse cohorts instead of work groups.

3.1 Dataset collection

The data was obtained by web crawling English online cryptocurrency news articles on all cryptocurrencies from various sources over a period of 42 months (July 2018–January 2022) using the cryptocurrency news aggregator CoinMarketCal website (CoinMarketCal 2022) resulting in a dataset with 2683 news.

3.2 Annotator selection

A group of 83 annotators participated in the annotation of CryptoLin. These annotators were all master's students at IE University with an average of eight years of work experience. The annotation took place during the course Finance, Risk and Fraud Analytics in 2022. Prior to the annotations, annotators had completed the Natural Language Processing course and learned necessary concepts around trading and cryptocurrency in the Finance, Risk and Fraud Analytics course.

Table 1 shows the count and percentage of students in three different categories: Blended, Fulltime-cohort-1, and Fulltime-cohort-2. The count column shows the number of students in each category while the percentage column shows the percentage of students in each category out of the total number of students.

For example, there were 26 students in the Blended category, an online version of the course where students are taking their courses in their own country, which represents 31.33% of all students. Similarly, there were 27 students in Fulltime-cohort-1 which represents 32.53% of all students and there were 30 students in Fulltime-cohort-2 which represents 36.14% of all students, the last two are formed from international students taking the course physically in Madrid, Spain.

Table 2 shows the annotator's count and percentage by nationality. Although Spanish and Germans are the highest numbers of students with 21.69% and 12.05% respectively, we can observe a very good diversity of nationalities with aims to provide less bias in the annotations process.

Table 3 shows the annotator's count and percentage by educational background. Although Business Administration and MBA are the highest numbers

Table 1 Annotators by cohort (finance, risk and fraud analytics course group)

Row labels	Count	Percentage (%)
Blended	26	31.33
Fulltime-cohort-1	27	32.53
Fulltime-cohort-2	30	36.14

Table 2 Annotators by nationality

Row labels	Count	Percentage (%)
American	2	2.41
Argentinian	2	2.41
Belgian	2	2.41
Canadian	1	1.20
Chilean	1	1.20
Chinese	2	2.41
Colombian	6	7.23
Costa Rican	1	1.20
Dominican	3	3.61
Dutch	1	1.20
Ecuadorian	3	3.61
French	4	4.82
German	10	12.05
Greek	1	1.20
Indian	2	2.41
Italian	5	6.02
Kazakhstani	1	1.20
Kenyan	1	1.20
Lebanese	2	2.41
Mexican	3	3.61
Panamanian	1	1.20
Filipino	1	1.20
Portuguese	2	2.41
Saudi Arabian	2	2.41
South African	1	1.20
Spanish	18	21.69
Swiss	3	3.61
Syrian	1	1.20
Thai	1	1.20

The table has three columns: Row Labels, count, and percentage. The first column lists the nationalities of the people in the group. The second column shows how many people belong to each nationality. The third column shows what percentage of the total number of people in the group each nationality represents

of students with 18.07% and 12.05% respectively, we can observe a very good diversity of educational backgrounds with aims to provide less bias in the annotations process.

The courses taken in Natural Language Processing, Finance, and Cryptocurrency provided the necessary knowledge for the topic. This, combined with the diversity of the students' countries and educational backgrounds, makes these 83 students ideal for the annotations task.

Table 3 Annotators by educational backgrounds

Row labels	Count	Percentage (%)
Accounting	3	3.61
Aeronautics	1	1.20
Arts	4	4.82
Biology	1	1.20
Business administration	15	18.07
Business engineering	4	4.82
Chemical engineering	1	1.20
Civil engineering	2	2.41
Commerce	1	1.20
Commercial engineering	1	1.20
Computer engineering	1	1.20
Computer science	8	9.64
Economics	7	8.43
Engineering	1	1.20
Finance	3	3.61
Human resource	1	1.20
Industrial engineering	3	3.61
International business	1	1.20
Law	1	1.20
Marketing	2	2.41
MBA / master in management	10	12.05
Politics	1	1.20
Science	7	8.43
Social science	1	1.20
Teaching	2	2.41
Tourism	1	1.20

The table has three columns: Row Labels, count, and percentage. The first column lists the education of the people in the group. The second column shows how many people belong to each educational background. The third column shows what percentage of the total number of people in the group each educational background represents

3.3 Annotation definitions and procedure

Each news was randomly assigned to three annotators, one from each cohort: Blended, Fulltime-cohort-1, and Fulltime-cohort-2. Meaning that each news would be annotated three times and consensus mechanism would be applied. No annotator had access to information of which news was assigned to whom. Annotators were asked to label each news with 1 if positive for the Cryptocurrency industry, 0 if neutral for the Cryptocurrency industry and -1 if negative for the Cryptocurrency

industry, no further guideline was given keeping the existing bias from the student and avoiding standardising biases.

Table 4 shows a pivoted version of the first row of the CryptoLin dataset selecting only the first five and last 3 columns of the dataset. Here one can observe the consensus 1 (positive) being achieved from assessed sentiments 1, 0 and 1 from randomly assigned annotators id=6, id=23 and id=17 respectively. Individual annotation labels (manual_labelling_1, manual_labelling_2 and manual_labelling_3) were included in the dataset, as a consensus is not always desirable and machine learning models can potentially perform better using original (even if contradictory) labels. This approach is supported by the findings of Kralj Novak et al. (2022), who argue that acknowledging multiple perspectives in the production of ground truth can improve the performance of machine learning models.

Table 5 presents the consensus mechanism used. First three columns shows the possible combination of the individual decisions taken by each the annotator and the decision column shows the final manual labelling assigned for the given combination. Column reasoning gives an explanation on the why the combined decision. Bold represents easy decision, italic and bolditalic represents decisions that have some sort of disagreement between annotators, but instead of using a 4th annotator in controversial cases as done by Daudert (2022), we decided to set the annotation equal to 0 (neutral) as the number of cases is small and, as we will see later, both inter-annotator agreement study and pre-trained sentiment analysis show good quality of the annotation as it is.

Table 6 shows a frequency table of final_manual_labelling column generate out of the CrypLin.csv dataset available in the project's Github Gadi and Ángel Sicilia (2022). In this table we can observe that 313, 1134 and 1236 news have been labeled negative, neutral and positive respectively after the three annotators consensus rule is applied (rule available at Table 5). This absolute frequency represents respectively 11.666%, 42.266% and 46.068% out of the total of 2683 news in the full dataset.

Table 4 Table containing the first news extracted from CoinMarketCal website (CoinMarketCal 2022), final_manual_labelling refers to the consensus of sentiments 1, 1 and 1 from randomly assigned annotators 22, 71 and 17 respectively

Key	Value
Id	0
Date	1/25/2022
News	Ripple announces stock buyback, nabs \$15 billion valuation
Final_manual_labelling	1
Text_span	{annotator1_id:22;annotator1_label:1 annotator2_id:71; annotator2_label:1 annotator3_id:17;annotator3_label:1 }
Manual_labelling_1	1
Manual_labelling_2	1
Manual_labelling_3	1

Table 5 Consensus table

Annotator 1	Annotator 2	Annotator 3	Decision	Reasoning	Count
-1	-1	-1	-1	Complete majority	380
-1	-1	0	-1	Majority with no objection	3
-1	-1	1	0	Minority report, majority with one objection	2
-1	0	-1	-1	Majority with no objection	0
-1	0	0	0	Majority	6
-1	0	1	0	Total disagreement	0
-1	1	-1	0	Minority report, majority with one objection	1
-1	1	0	0	Total disagreement	2
-1	1	1	0	Minority report, majority with one objection	2
0	-1	-1	-1	Majority with no objection	2
0	-1	0	0	Majority	3
0	-1	1	0	Total disagreement	3
0	0	-1	0	Majority	6
0	0	0	0	Complete majority	864
0	0	1	0	Majority	13
0	1	-1	0	Total disagreement	1
0	1	0	0	Majority	13
0	1	1	1	Majority with no objection	17
1	-1	-1	0	Minority report, majority with one objection	2
1	-1	0	0	Total disagreement	3
1	-1	1	0	Minority report, majority with one objection	2
1	0	-1	0	Total disagreement	1
1	0	0	0	Majority	15
1	0	1	1	Majority with no objection	21
1	1	-1	0	Minority report, majority with one objection	3
1	1	0	1	Majority with no objection	13
1	1	1	1	Complete majority	1305
					2683

Column decision indicates the final_manual_labelling assigned depending on annotation given by annotator 1, 2 and 3, Column reasoning gives an explanation of the decision. Note minority report to default the labelling to neutral in case of one objection is used. Count represents the number of occurrences in the dataset

Table 6 Summary frequency table of final_manual_labelling

Final_manual_labelling	Count	% of total
-1 (negative sentiment)	385	14.35
0 (neutral sentiment)	942	35.11
1 (positive sentiment)	1356	50.54
Total	2683	100.00

4 Inter-annotator agreement study

4.1 Agreement measures

As a first quality control of CryptoLin we used the steps followed by Jacobs and Veronique (2022) and calculated the following inter-rater reliability coefficients: Fleiss' kappa (κ) (Fleiss 1971), Krippendorff's alpha (α) (Krippendorff 2006), and Gwet's AC1 (Gwet 2008) on the aligned span annotations using the Multi Class Confusion Matrix Library for Python provided by Haghighi et al. (2018).

Table 7 presents Fleiss's Kappa, Krippendorff's alpha and Gwet's AC1 inter-rater reliability coefficients results, measured between annotators 1 and 2 [Coeff (1–2)], between annotators 1 and 3 [Coeff (1–3)], and between annotators 2 and 3 [Coeff (2–3)], and between the 3 annotators [Coeff (1–2–3)]. In addition, it provides the interpretation of Kappa metric using Landis-Koch, Fleiss, Altman, Cicchetti benchmarks in the last four rows. Details on Kappa's range and the interpretation for such benchmarks can be found at 10 in the Appendix. As one can observe from the same table, Inter-rater reliability coefficient Fleiss's Kappa presents Almost Perfect, Excellent, Very Good and Excellent inter-annotator agreement according to Landis-Koch, Fleiss, Altman, Cicchetti benchmarks respectively for all combinations of annotators, demonstrating an acceptable inter-annotator agreement among the annotators. Table 7 provides details of several Krippendorff's alphas. Krippendorff's alpha is a statistical measure of agreement that offers several advantages. It has multiple versions, depending on the type of variable used, including binary, nominal, ordinal, interval, ratio, polar, and circular metrics. Additionally, it can be applied to any number of annotators. In our study, our labeling is ordinal (negative, neutral, and positive) and we have three annotators. As such, the Ordinal 3-annotators Krippendorff's alpha of 0.952 is the most appropriate measure to use and it indicates a very good inter-annotator agreement. We have included other versions of

Table 7 Fleiss's Kappa, Krippendorff's Alpha and Gwet's AC1 inter-rater reliability coefficients for annotators 1 and 2 (Coeff (1–2)), 1 and 3 (Coeff (1–3)), 2 and 3 (Coeff (2–3)), and for the 3 annotators (Coeff (1–2–3)) as well as its benchmark interpretation according to Landis and Koch (1977), Fleiss (1971), Altman (1990) and Cicchetti (1994) benchmarks. Krippendorff 2 and 3 annotators nominal and ordinal alpha is also presented

Metric	Coeff (1–2)	Coeff (1–3)	Coeff (2–3)	Coeff (1–2–3)
Fleiss' κ	0.942	0.942	0.944	0.943
Kappa's StdErr	0.006	0.006	0.006	0.008
Kappa's 95% C.I	(0.958, 0.972)	(0.958, 0.972)	(0.96, 0.973)	(0.926, 0.959)
Nominal Krippendorff's α	0.942	0.942	0.944	0.943
Ordinal Krippendorff's α	0.950	0.953	0.952	0.952
Gwet's AC1	0.9499	0.9499	0.952	0.9499
Landis and Koch (1977) benchmark	Almost perfect	Almost perfect	Almost perfect	Almost perfect
Fleiss (1971) benchmark	Excellent	Excellent	Excellent	Excellent
Altman (1990) benchmark	Very good	Very good	Very good	Very good
Cicchetti (1994) benchmark	Excellent	Excellent	Excellent	Excellent

Krippendorff's alpha for researchers who may wish to investigate this further. Our findings do not contradict those of Mozetič et al. (2016), which demonstrated that nominal inter-annotator agreement can be improved. In the case of Mozetič et al. (2016), using interval Alpha yields higher agreement, while in our case using the correct choice of Ordinal Krippendorff's alpha yields higher agreement.

5 Further enrichment and performance analysis

5.1 Enriching the dataset with an event study flag for future event studies

In order to enrich the dataset for future event studies, we decided to provide the field `type_abnormal_return_fama_frech` in the dataset.

An event study is a statistical method used to assess the impact of an event on the value of a security. An illustration of an event study is the examination of the stock price on the ex-dividend date. If the market operates efficiently, then, on average, the stock price decreases by the dividend amount. For example, consider Hindustan Zinc (HZL), which declared a dividend of Rs 23.5 per share with a record date of July 14, 2023[2]. The ex-dividend date for this stock was July 13, 2023 (Zee-Biz WebTeam 2023), indicating that any individual who purchased the stock on or after this date would not be entitled to receive the dividend; instead, the seller would receive it. On July 12, 2023, the closing price of HZL was Rs 327.1 (CFI 2023). On July 13, 2023, the opening price of HZL was Rs 303.6[2], lower than the closing price by Rs 23.5, precisely the dividend amount. This observation suggests that the market adjusted the stock price for the dividend payment and that HZL was trading ex-dividend on July 13, 2023.

The methodology chosen for the construction of the type abnormal return fama frech field involves using the Fama French Three Factor Model to obtain the excess return within an asset. This model considers the impact of capitalization and book value in returns of investments by adding a size and a value risk factor to the original CAPM market risk component (Teo 2021). A support dataset was obtained from Dartmouth's website (French 2022), which provided the Market Factor, Size Factor, Value Factor, and Risk Free Rate for the Fama and French Three Factor Model (Teo 2021). This can be seen in Table 8, where data is ordered according to dates.

The construction of an abnormal return field using the Fama-French model involves several decisions regarding parameter selection. While this field may serve as a starting point, it may be necessary for other researchers to adjust the abnormal return field to suit their specific needs.

- `Type_abnormal_return_fama_frech = 1` indicates an day with positive abnormal return in one of the following 4 days.
- `Type_abnormal_return_fama_frech = 0` indicates expected return in one of the following 4 days either positive or negative.
- `Type_abnormal_return_fama_frech = -1` indicates an day with negative abnormal return in one of the following 4 days.

Table 8 Market factor, size factor, value factor, and risk free rate for the Fama and French three factor model from website (French 2022)

Date	Mkt-RF	SMB	HML	RF
2018-07-30	- 0.7	- 0.22	1.57	0.008
2018-07-31	0.51	0.85	- 1.11	0.008
2018-08-01	- 0.13	0.06	- 0.19	0.007
2018-08-02	0.67	0.32	- 0.67	0.007
2018-08-03	0.31	- 1.07	0.51	0.007
2018-08-06	0.46	0.25	- 0.33	0.007
2018-08-07	0.29	0.03	- 0.16	0.007
2018-08-08	- 0.04	- 0.12	0.26	0.007
2018-08-09	- 0.05	0.37	- 0.36	0.007
2018-08-10	- 0.6	0.39	- 0.21	0.007
2018-08-13	- 0.46	- 0.14	- 0.3	0.007
2018-08-14	0.69	0.3	0.21	0.007
2018-08-15	- 0.91	- 0.54	- 0.12	0.007
2018-08-16	0.86	- 0.07	0.26	0.007
2018-08-17	0.3	0.09	0.02	0.007
2018-08-20	0.25	0.12	0.14	0.007
2018-08-21	0.33	0.9	0.12	0.007
2018-08-22	0.05	0.37	- 0.35	0.007
2018-08-23	- 0.19	- 0.06	- 0.33	0.007
2018-08-24	0.62	- 0.02	- 0.57	0.007
2018-08-27	0.74	- 0.68	- 0.17	0.007
2018-08-28	- 0.01	- 0.12	- 0.28	0.007
2018-08-29	0.56	- 0.04	- 0.57	0.007
2018-08-30	- 0.41	0.28	- 0.42	0.007
2018-08-31	0.08	0.5	- 0.39	0.007
2018-09-04	- 0.11	- 0.38	0	0.008
2018-09-05	- 0.41	- 0.17	0.66	0.008
2018-09-06	- 0.44	- 0.35	- 0.24	0.008
2018-09-07	- 0.18	0.1	- 0.21	0.008
2018-09-10	0.23	0.12	- 0.37	0.008
:	:	:	:	:
:	:	:	:	:
2021-12-27	1.22	- 0.09	0.28	0

The methodology employed involves the utilization of the Fama-French Three Factor Model to determine the excess return of an asset. To capture the full effect of the selected days, a 7 day time frame was implemented to allow for comparison of daily valuations. This range encompasses the release day, 2 days prior, and 4 days following the release of the news. The selection of a 7 day time frame, including the release day, 2 days prior, and 4 days following the release of the news, facilitates a more thorough analysis of the event's impact on the asset's returns. This time frame provides sufficient data points to capture any immediate or delayed responses to the news while remaining brief enough to minimize the influence of external factors. To

achieve approximately 5% abnormal days (Investopedia, 2024), abnormal returns are calculated as the difference between actual and expected returns and are considered positive abnormal if they exceed 180 times the expected return or negative abnormal if they fall below -180 times the expected return.

5.2 Further calculated fields and Performance analysis

In order to assess the practical quality of the CryptoLin dataset, we decided to include the following list of pre-trained NLP algorithms.

- Vader sentiment analysis (Hutto and Gilbert 2014),
- TextBlob sentiment analysis (Loria 2013),
- Flair NLP library (Akbik 2019),
- FinBERT financial sentiment analysis with BERT (Liu 2020), for FinBERT we include the 3 predictions—finbert_positive, finbert_negative and finbert_neutral scores.

Table 9 shows a comparison of 4 pre-trained algorithms Vader, TextBlob, Flair and FinBERT using each of its predictions. For all algorithms, we observed that the current classes assigned by the algorithms were performing poorly. As a result, we decided to use the probabilities to define optimal cut-offs for a 3-class labeling scheme, with -1 , 0 , and 1 representing negative, neutral, and positive, respectively. In the case of FinBERT, using the common approach of selecting the class with the highest probability among the negative, neutral, and positive classes also resulted in poor performance. Therefore, we decided to analyze each probability separately. We applied roc curve method (from sklearn.metrics) to find out the two best threshold

Table 9 Comparison of 4 pre-trained algorithms in CryptoLin

Sentiment algorithm	Best negative threshold	Best positive threshold	Accuracy (%)	F1-score (weighted) (%)	Krippendorff's alpha (%)
Vader	0.0258	0.0258	44.2	43.62	17.88
Textblob	0.00568*	0.0057	26.91	23.9	-13.91
Flair	0.94471*	0.9447	23.56	21.97	-22.88
Finbert_positive	0.08026	0.0803	54.98	51.2	47.48
Finbert_negative	-0.01521	-0.0152	58.93	57.66	49.79
Finbert_neutral	0.68332	0.6833	38.54	33.15	2.76

TextBlob and Flair show poor performance for 3-class classification. Vader and FinBERT show reasonable performance in all metrics (Accuracy, in both F1-scores and Krippendorff's alpha) demonstrating data was not annotated randomly, in other words, the labelling is useful. FinBERT (negative) presents best performance indicating advantage of its expertise in the financial field

*Means that no neutral found. The Krippendorff's Alpha values in the table represent the agreement between the sentiment model (algorithm) predictions and the ground truth labels, not between different human raters

for each pre-trained algorithm in order to turn the sentiment score into class -1 (negative), 0 (neutral) and 1 (positive). For doing that, we created two target variables `y_else_or_negative` and `y_positive_or_else`. Next, we applied roc curve method (from `sklearn.metrics`) which returns the best threshold for each algorithm and the target variables `y_else_or_negative` and `y_positive_or_else` resulting into the `best_negative_threshold` and `best_positive_threshold` respectively.

We included the 3-class accuracy, as well the 3-class F1-score using the weighted average options which calculates the weighted mean of the F1 scores for each class, where each class's score is weighted by its prevalence which deems more indicated for imbalance classes. We included 3-class Krippendorff's alpha, which ranges from -1 to 1. A value of 1 indicates perfect agreement among raters, a value of 0 indicates agreement that is no better than chance, and negative values indicate systematic disagreement among raters.

According to Table 9, FinBERT with the negative threshold of -0.01521 has the highest accuracy (58.93%) and macro F1-score (57.66%) among all the algorithms. It also has the highest Krippendorff's alpha (49.79%), indicating good agreement among raters. FinBERT (negative) presented the best performance indicating advantage of its expertise in the financial field for further fine-tuning of sentiment analysis algorithms. On the other hand, Flair has the lowest accuracy (23.56%), macro F1-score (21.97%), and Krippendorff's alpha (-22.88%), indicating poor performance and systematic disagreement among raters.

6 Conclusions

In this research, we introduce the Cryptocurrency Linguo (CryptoLin), an innovative corpus encompassing more than three years of cryptocurrency news. Each article within CryptoLin has been meticulously annotated by human evaluators with a label of -1.0, 0.0, or 1.0 to signify whether the news is negative, neutral, or positive, respectively. Three individuals from distinct cohorts were randomly assigned to annotate each article utilizing a majority voting methodology to establish the final label through consensus. In instances of pronounced disagreement, the default label was designated as neutral.

CryptoLin comprises several components, including an id, date, news, and `final_manual_labelling` with consensus annotation. Additionally, it features a `text_span` that provides details on the labeling process. The corpus also incorporates four pre-trained sentiment algorithms: Vader, Textblob, Flair, and FinBERT (positive, negative, and neutral).

The inter-rater reliability coefficients Fleiss's Kappa, Krippendorff's Alpha, and Gwet's AC1 all demonstrate an acceptable level of inter-annotator agreement.

A preliminary analysis utilizing Vader, Textblob, Flair, and FinBERT confirmed the utility of the dataset labelling for further refinement of sentiment analysis algorithms. Notably, FinBERT (negative) exhibited the best performance, indicating that an algorithm specifically trained to analyze the sentiment of financial content is necessary.

The rules governing the selection and assignment of annotators for CryptoLin appear to effectively mitigate typical cohort biases and thus present a valuable opportunity for further research into annotator selection, assignment, and biases. Consequently, CryptoLin enhances current understanding by providing a robustly labelled dataset that affords researchers the opportunity to expand studies in the field of sentiment algorithms and NLP in general using cryptocurrency news events.

Appendix

Table 10 shows the benchmarks for interpreting the kappa coefficient which is used to assess inter-rater reliability or agreement. The kappa coefficient ranges from -1 to 1 and measures the agreement between two raters who are rating the same thing. The table shows four benchmarks for interpreting kappa values: Landis and Koch (1977) benchmark, Fleiss (1971) benchmark, Altman (1990) benchmark, and Cicchetti (1994) benchmark. Each benchmark has its own range of values and interpretation of those values. For example, according to Landis-Koch benchmark, a kappa

Table 10 Interpretation of Kappa metric using number with Landis-Koch, Fleiss, Altman, Cicchetti benchmarks

Landis and Koch (1977) benchmark range	Landis-Koch benchmark interpretation
$0.00 \leq \kappa < 0.2$	Slight
$0.20 \leq \kappa < 0.4$	Fair
$0.40 \leq \kappa < 0.6$	Moderate
$0.60 \leq \kappa < 0.8$	Substantial
$0.80 \leq \kappa \leq 1.0$	Almost Perfect
Fleiss (1971) benchmark range	Fleiss benchmark interpretation
$\kappa < 0.4$	Poor
$0.40 \leq \kappa < 0.75$	Intermediate to Good
$0.75 \leq \kappa \leq 1.0$	Excellent
Altman (1990) benchmark range	Altman benchmark interpretation
$0.00 \leq \kappa < 0.2$	Poor
$0.20 \leq \kappa < 0.4$	Fair
$0.40 \leq \kappa < 0.6$	Moderate
$0.60 \leq \kappa < 0.8$	Good
$0.80 \leq \kappa \leq 1.0$	Very Good
Cicchetti (1994) benchmark range	Cicchetti benchmark interpretation
$0.00 \leq \kappa < 0.4$	Poor
$0.40 \leq \kappa < 0.59$	Fair
$0.59 \leq \kappa < 0.74$	Good
$0.74 \leq \kappa \leq 1.0$	Excellent

value between 0.00 and 0.20 indicates slight agreement while a value between 0.80 and 1.00 indicates almost perfect agreement.

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Data availability Both the CryptoLin dataset and the Jupyter Notebook with the analysis, for reproducibility purposes, are available at the project's Github [?] - <https://github.com/manoelgadi/CryptoLin>

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